

Quasar Clustering Using Machine Learning

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I. ABSTRACT

We used K-Means, Density-Based Spatial Clustering of Applications with Noise (DBSCAN), and Gaussian Mixture Model (GMM) algorithms to study quasar clustering between redshifts 3 and 6 with data from the Sloan Digital Sky Survey. Our analysis shows a lack of evidence of clustering, possibly indicating that the large-scale structure of the early universe is more uniform than previously thought. This could lead to further research into quasar clustering using a combination of physical and spatial properties.

II. INTRODUCTION

Quasars are a type of Active Galactic Nuclei (AGNs) which are extremely bright centers of galaxies that are powered by a supermassive black hole consuming material. They were discovered when astronomers began to identify radio sources of objects outside of the Milky Way Galaxy. Some of the radio sources coincided with unusually bright, blue stars with fuzzy halos that were later classified as "quasi-stellar radio sources," or "Quasars" for short. Quasars are approximately 100 times more luminous than other celestial objects discovered at the same distance. Their luminosity can vary dramatically, sometimes within days, indicating their relative compactness for the amount of light the object produce, leading to extremely high radiation pressure. Due to this pressure, a quasar must have a mass of at least one million solar masses to prevent an explosion from its own radiation (Cooper, 2023).

Researchers theorize that Quasars eventually "run out of fuel" and behave more like galaxies, making Quasars a stage in the galaxy evolutionary process. By extending upon this theory, researchers believe that if Quasars are a stage in galactic evolution, then Quasar clustering should exist if galaxy clustering exists (Galaxy Evolution, 2024). Quasars are theorized to be embedded in the evolving cosmic web of dark matter (DM) halos that form the backbone of all structures in the universe (Martini & Weinberg, 2001). Quasar luminosity can be used to study these DM halo masses to infer values for quasar duty cycles and study quasar lifetimes, and halo mergers. Quasars are also used as cosmological tracers as they are a stage in galactic evolution and can be used to study galactic properties and evolution (Martini & Weinberg, 2001). Quasar clustering provides crucial information about the large-scale structure of the universe by shedding further light on the hierarchical structure formation theory of the universe, especially on the hierarchical formation of quasars and galaxies.

III. LITERATURE REVIEW

There have been numerous past research conducted on Quasar clustering, but most papers focused on lower redshifts ($z < 3$), while only a select focused on higher redshifts ($z \geq 3$). A past study conducted by Jose Ordonez and Carolyn Stripling utilized two clustering algorithms, Nearest Neighbors (NN) and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to analyze approximately 17,000 Quasars obtained from the Sloan Digital Sky Survey Database (SDSS). With the NN algorithm, Ordonez and Stripling calculated an average Quasar-Quasar distance of 41.8 Mpc. After, they used this distance to implement DBSCAN with a 'min_samples' of 5. They discovered that Quasar Clustering exists up to redshifts 2.2, except for between $z = 0.5$ and $z = 1$. There was an especially high concentration of Quasar clusters within $z < 0.5$. Ordonez and Stripling concluded that there is evidence of Quasar Clustering at lower redshifts instead of high redshifts due to Quasars and their host galaxies needing time to interact gravitationally to cluster in higher-density areas (2022).

In another study, researchers sought to demonstrate that Quasars are useful tracers in the large-scale universe because understanding Quasar clustering is the key to understanding cosmic evolution and dark matter distribution. They wanted to quantify Quasar bias; in other words, compare Quasar clusters to the underlying dark matter distribution. From the Extended Baryon Oscillation Spectroscopic Survey (eBOSS) database, researchers studied 70,000 Quasars with $0.9 \leq z \leq 2.2$ (Laurent et al., 2017). To correct potential errors that can bias Quasar measurements, the team accounted for survey geometry as artificial clustering signals can arise from the survey design and variations in the observed conditions. Lastly, the group recognized that the same selection might introduce some bias leading to their decision to incorporate many data points. Laurent et al. incorporated a two-point correlation function and a power spectrum to analyze Quasar clusters as these are the conventional methods of measuring clusters. By accounting for redshift space corrections, bias determination, and systematic errors, the team found Quasar Bias within their sample and that this value increased with redshift, hypothesizing that Quasars clustered more at higher redshifts (Laurent et al., 2017).

While the studies above focused on lower redshifts, there has been previous research done at higher redshifts. One team from Cornell University studied Quasar and galaxy clustering at $z = 6$. This group collected data from the EIGER JWST program with $2 \leq z \leq 6$. After building an empirical Quasar population model, the team concluded that there is strong ev-

idence of evolution of the host halo mass and duty cycle of Quasars at $2 \leq z \leq 6$ indicating that there is evidence of more Quasar clustering at higher redshifts (Pizzati et al, 2024). Another team focused on Quasar clustering at $z = 6$, but with a much smaller sample size of just 173 Quasars with $z \geq 5.6$. By using imaging observations and comparing their results with internal simulations, the team concluded that while they only had very few pairs of Quasar clustering at $z = 6$, they expect to see more Quasar clustering systems at higher redshifts due to their application of the luminosity function to their Quasar samples (Greiner et al., 2021).

Overall from our literature review, Quasar clustering is said to exist at both low and high redshifts. Quasar clustering at lower redshifts are already a fundamental ingredient on the current scientific world’s understanding of Supermassive Black Holes (SMBHs), their accretion mechanisms, and the co-evolution with their host galaxies. Even though Quasar clustering is challenging to study at higher redshifts due to their great distances, it is highly significant in studying Quasar formation history, Quasar duty cycle (ratio of its activity and inactivity time), Quasar lifetimes, and SMBHs’ properties. While there is current consensus among low redshift clustering models, there is very little to no consensus among high redshift models. From this, future research can focus on Quasar clustering towards higher redshift range, especially within $3 \leq z \leq 6$ as there has been very few published research studies focusing within this range with any significant conclusions, which is what our group has decided to focus in this paper.

IV. COMPUTER ANALYSES

A. Data Cleaning

To obtain sufficient and relatively accurate data, we decided to use the Sloan Digital Sky Survey (SDSS) as it has a large collection of quasars (SDSS, 2023). When analyzing our data, we wanted to ensure to only use clearly defined quasars because sometimes galaxies or other objects could be misclassified as quasars based on physical properties which can cause inaccurate results. To accomplish this, we decided to filter our data according to specific color filter magnitude ranges. We learned that most quasars’ r filter magnitudes are between 17.25 and 22.75 inclusive (Carrasco et al., 2015). Additionally, most quasars with redshifts from three to four tended to have $g-r$ magnitudes between 0.5 and 1.5, $r-i$ magnitudes between 0 and 0.5, and $i-z$ magnitudes between 0.5 and 1.0. For Quasars with redshifts greater than four, their $r-i$ magnitudes are typically between 0.5 and 2.0, and $i-z$ magnitudes between 1.0 and 2.5 (Richards et al., 2007). Lastly, we found that most Quasars fall between 18 and 21 on the g color filter magnitude (Hughes et al., 2022). After filtering our dataset, we had 13489 quasars remaining.

We also decided to convert the celestial coordinates, RA and DEC, to galactic longitude and latitude for better representation and understanding of each individual point. Additionally, we decided to create a third axis by using the redshift

as a distance indicator in Mpc and creating a new “Distance” column using Hubble’s Law. To obtain the most accurate results, we used the most widely accepted value for Hubble’s Constant, 72 km/s/Mpc (Mukhanov, 2005). When plotting our original plot, we noticed an abnormal smooth edge along the majority of quasars. To account for this, we performed a Gaussian kernel transform to identify non-linear relationships and map them as linear ones. When analyzing the distribution of quasars in each redshift range, we noticed a heavily skewed distribution towards lower redshifts. To address this, we decided to cluster our quasars using their redshifts as weights by multiplying redshift to the latitude, longitude, and distance columns. By doing this, quasars that are at higher redshifts would have heavier weights making them more significant. Lastly, because due to the extreme scale differences for the latitude, longitude, and distance columns, we decided to standardize each one to obtain more accurate results.

B. Machine Learning

After the data cleaning process, we built three different machine learning models to find any clustering systems. Firstly, we used the K-Means clustering algorithm. To implement this model, we had to determine the optimal number of clusters. We used the Elbow Method and Silhouette Score Analysis Method in which both algorithms returned six optimal clusters. When analyzing our results, we found that the average cluster radius was approximately 0.733.

Subsequently, we created the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) model. As part of the DBSCAN algorithm, for our optimal ϵ value, the maximum distance two points are considered neighbors of each other, we used the KNN (K-Nearest Neighbor) algorithm to create a K-Distance Graph which returned an optimal ϵ value of 0.732. Additionally, we also had to determine the minimum number of quasars a cluster should contain in order for it to be considered an actual cluster. We chose four as an optimal number as previous research papers chose numbers close to ours and we wanted to increase our chances of getting clustering evidence. Afterwards, our algorithm returned one cluster while identifying 22 outliers. When analyzing our results, we found that the average cluster radius was approximately 7.53.

Lastly, we used the Gaussian Mixture Model (GMM) to determine if it will yield better results. We calculated the Bayesian Information Criterion (BIC) for a range of number of clusters to determine the optimal number. The BIC is a statistical measure used to compare different statistical models based on their fit to the data while penalizing for the number of parameters, essentially favoring models that best explain the data with the fewest parameters which helps prevent overfitting. After analyzing the different BIC values, we concluded that 24 was the optimal number as it had the lowest BIC value. When analyzing our results, the average cluster radius was approximately 0.333. It is important to note that there are no units for all of the cluster radii as we clustered using the standardized values.

V. RESULTS

When analyzing each machine learning model's results, we see that there are inconsistencies among all models. The graphs below show a scatterplot of quasars plotted based on their galactic longitudes and latitudes. Then, they are color coded based on the cluster assigned. We see that each model does not return a similar number of clusters nor are there any overlapping clusters (FIG. 1).

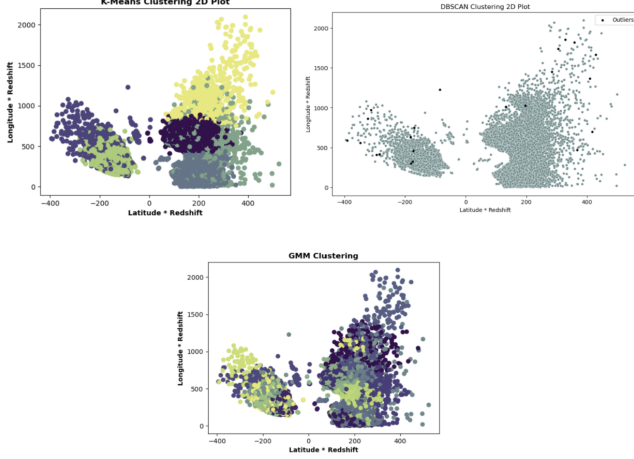


FIG. 1. Visualizations of K-Means, DBSCAN, GMM Clustering Algorithms Results

Because there is a great inconsistency in the number of clusters and each cluster's position, it is hard to conclude that there is significant evidence of quasar clustering within this redshift range. When we also analyze the number of quasars in each cluster, the number fluctuates greatly (FIG. 2). Each graph shows the distribution of the number of quasars for each cluster.

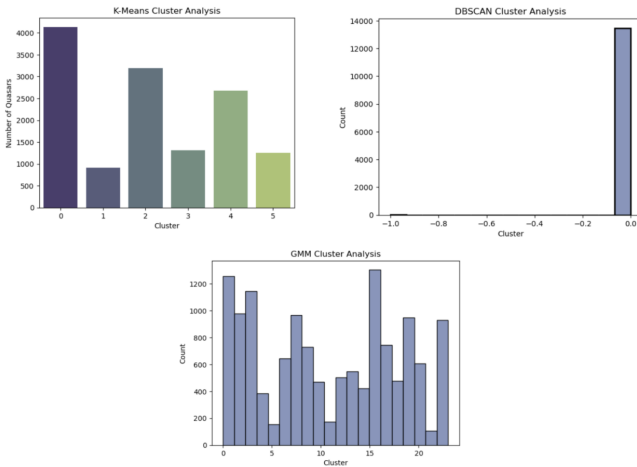


FIG. 2. Algorithms Analysis Plots

With the K-Means and GMM algorithms, there is a relatively weak even distribution. However, the DBSCAN algo-

rithm only returns one cluster that contains most of the quasars except for a few outliers.

VI. DISCUSSION

Overall, due to strong inconsistencies, our research does not point toward evidence of clustering. Possible reasons for these inconsistencies could be attributed to our data set. For example, when examining the 2D plots of each of our models, we see that there is a very large missing gap between latitudes -20 and 20. The main reason for this could be the Milky Way band in which the heavy dust prevented further quasars from being detected.

In a study conducted on 30,239 quasars with redshifts from 0.3 to 2.2, they found a predictable clustering pattern in quasars at these redshifts based on spatial properties (Ross et al., 2009). They found a clustering correlation length of about 5.95 Mpc/h, which means that quasars separated by that distance were twice as likely to be clustered together in a random distribution. This number indicates that quasars tend to be in somewhat dense groups that could be considered clusters but do not exhibit extremely strong clustering. That goes along with the fact that some of our clustering algorithms found very dense areas of quasars but not necessarily quasar clustering. They found a very stable and predictable pattern in lower redshift quasars (Ross et al., 2009). Another thing that they found is that the clustering strength of the quasars remained relatively constant across the redshift range. This could indicate that quasars have been clustering in similar ways for a significant period, but we did not see this same pattern in the older quasars that we surveyed because our quasars were still more dispersed than what they were finding. They did find significant clustering within scales of 1 to 25 Mpc/h with clustering diminishing at larger distances using a two-point correlation function. They found no significant clustering at distances greater than 100 Mpc/h. This all suggests that quasars do not exhibit clustering on very large cosmic scales (Ross et al., 2009).

In another study conducted on quasars with redshifts greater than 2.9, they found that there was very strong clustering in this redshift range. More specifically, quasars with redshifts greater than 3.5 were more clustered than those between 2.9 and 3.5. This study found a clustering correlation length of 16.9 Mpc/h for the lower range and a clustering correlation length of 24.3 Mpc/h for the higher redshifts. This implies extremely strong clustering of quasars in these redshift ranges, especially as redshift increases. They attributed this clustering to the belief that the universe was more densely populated (Shen et al., 2007). This is a vastly different result than what our data showed. Our data showed moderate to little clustering in quasars with redshifts from three to six. Additionally, our results show that quasars seem to be more dispersed or congregated in densely populated areas throughout the universe as opposed to strongly clustered. The differences in detected clusters could be attributed to the differing number of samples utilized and due to our choice to weight quasars by their redshift.

VII. CONCLUSION

Overall, our research was able to answer our original research topic of trying to find quasar clustering systems between redshifts 3 and 6. Although our results and analyses pointed towards no significant evidence of clustering, we were still able to learn a lot more about the universe. Specifically, our results seem to indicate that the large-scale structure of the universe is more uniform than the early universe. Additionally, our results point toward future research about quasar clustering using different properties. Our research studied quasar clustering based on spatial properties (i.e. galactic longitude and latitude). However, future research could be done by studying the physical properties such as color filter magnitudes and trying to find evidence of clustering systems. From another perspective, our results could mean that there is no clustering of quasars at higher redshifts, which would contradict conclusions made by previous research (Shen et al., 2007). In the future, our research can be modified to study quasar clustering in smaller blocks of redshift ranges.

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AUTHOR DECLARATIONS

A. Conflict of Interest Statement

The authors have no conflicts to disclose.

B. Author Contributions

Anthony Yang: Conceptualization (equal); Data Curation (lead); Formal Analysis (lead); Methodology (lead); Software (lead); Writing - review and editing (lead); Writing - original

draft (lead). Mia Chaplin: Conceptualization (equal); Data Curation (supporting); Project Administration (lead); Visualization (supporting); Writing – review and editing (supporting); Writing – original draft (supporting). Prisha Desai: Conceptualization (equal); Data Curation (supporting); Project Administration (supporting); Visualization (lead); Writing – review and editing (supporting); Writing - original draft (supporting).

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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